

## YAN AN, China

WMO: 538450

Lat: 36.5757N

Lon: 109.4498E

Elev: 1181

StdP: 87.92

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -14.5      | -12.5 | -25.1                         | 0.4 | -7.1 | -22.8 | 0.6 | -7.1 | 5.7                   | -0.8 | 4.9 | -3.0 | 1.8                   | 220  | 0.415 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.3                   | 33.3            | 19.1 | 31.8 | 18.8 | 30.4 | 18.7 | 23.1                | 28.3 | 22.2 | 27.2 | 21.5 | 26.4 | 2.2                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 21.6                            | 18.9 | 25.6 | 20.9 | 18.0 | 24.8 | 20.0 | 17.0 | 24.3 | 75.5          | 28.5 | 72.1 | 27.4 | 68.9 | 26.0 | 25.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|-----|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |     | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%  | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c) | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 5.5  | 4.5 | 3.8                        | DB  | -18.1              | 35.5 | 2.2                                                | 1.7 | -19.7      | 36.8 | -21.0      | 37.8 | -22.2      | 38.7 | -23.7 | 40.0 |
| (5)               |      |     |                            | WB  | -18.9              | 24.2 | 2.0                                                | 0.8 | -20.3      | 24.8 | -21.5      | 25.3 | -22.6      | 25.8 | -24.1 | 26.4 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 11.2  | -4.2  | 0.0   | 6.6   | 13.5  | 18.1  | 22.2  | 24.1  | 22.6  | 17.6  | 11.3  | 4.2   | -2.4 | (6)  |
| (7)  |                                                                   | DBStd       | 10.33 | 3.61  | 4.27  | 4.47  | 4.26  | 3.36  | 2.68  | 2.48  | 2.59  | 3.28  | 3.36  | 3.76  | 3.52 | (7)  |
| (8)  |                                                                   | HDD10.0     | 1444  | 440   | 279   | 123   | 15    | 1     | 0     | 0     | 0     | 0     | 26    | 176   | 386  | (8)  |
| (9)  |                                                                   | HDD18.3     | 3114  | 698   | 512   | 364   | 153   | 45    | 3     | 0     | 2     | 51    | 218   | 423   | 644  | (9)  |
| (10) |                                                                   | CDD10.0     | 1881  | 0     | 0     | 18    | 120   | 253   | 366   | 438   | 390   | 228   | 66    | 2     | 0    | (10) |
| (11) |                                                                   | CDD18.3     | 509   | 0     | 0     | 0     | 8     | 39    | 119   | 179   | 134   | 29    | 0     | 0     | 0    | (11) |
| (12) |                                                                   | CDH23.3     | 4683  | 0     | 0     | 7     | 148   | 517   | 1202  | 1563  | 996   | 237   | 12    | 0     | 0    | (12) |
| (13) |                                                                   | CDH26.7     | 1683  | 0     | 0     | 1     | 38    | 147   | 476   | 638   | 319   | 64    | 1     | 0     | 0    | (13) |
| (14) | Wind                                                              | WSAvg       | 1.7   | 1.6   | 1.7   | 1.9   | 2.0   | 2.0   | 1.8   | 1.7   | 1.5   | 1.5   | 1.5   | 1.6   | 1.6  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 524   | 4     | 7     | 17    | 28    | 43    | 62    | 121   | 105   | 77    | 39    | 17    | 3    | (15) |
| (16) |                                                                   | PrecMax     | 860   | 14    | 25    | 54    | 83    | 116   | 171   | 485   | 192   | 174   | 82    | 61    | 17   | (16) |
| (17) |                                                                   | PrecMin     | 365   | 0     | 0     | 1     | 6     | 11    | 21    | 26    | 35    | 18    | 6     | 0     | 0    | (17) |
| (18) |                                                                   | PrecStd     | 108   | 4     | 6     | 13    | 17    | 25    | 33    | 80    | 46    | 39    | 24    | 17    | 4    | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 9.7   | 15.7  | 24.1  | 31.0  | 32.2  | 35.1  | 35.5  | 33.4  | 31.9  | 24.8  | 18.8  | 10.1 | (19) |
| (20) |                                                                   | MCWB        | 1.8   | 5.5   | 9.8   | 15.0  | 15.5  | 17.5  | 20.1  | 20.6  | 19.1  | 13.8  | 8.9   | 2.5   | 2.5  | (20) |
| (21) |                                                                   | 2%          | DB    | 6.9   | 12.0  | 20.2  | 27.2  | 30.0  | 32.9  | 33.6  | 31.5  | 28.3  | 22.2  | 16.2  | 7.7  | (21) |
| (22) |                                                                   | MCWB        | 0.3   | 3.4   | 8.4   | 12.8  | 14.7  | 17.5  | 20.1  | 20.6  | 18.1  | 12.6  | 7.5   | 1.1   | 1.1  | (22) |
| (23) |                                                                   | 5%          | DB    | 4.6   | 9.9   | 17.8  | 24.7  | 28.3  | 31.2  | 32.0  | 29.9  | 25.9  | 20.2  | 13.5  | 5.8  | (23) |
| (24) |                                                                   | MCWB        | -1.1  | 2.5   | 7.2   | 11.3  | 14.3  | 17.3  | 20.1  | 20.4  | 17.0  | 11.9  | 6.5   | 0.0   | 0.0  | (24) |
| (25) |                                                                   | 10%         | DB    | 2.5   | 7.6   | 15.2  | 22.3  | 26.4  | 29.6  | 30.3  | 28.4  | 23.8  | 17.9  | 11.2  | 3.8  | (25) |
| (26) |                                                                   | MCWB        | -2.3  | 1.5   | 6.0   | 10.3  | 13.7  | 16.9  | 20.2  | 20.1  | 16.4  | 10.8  | 5.6   | -1.0  | -1.0 | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 2.3   | 6.1   | 11.2  | 16.2  | 18.2  | 21.6  | 24.2  | 23.9  | 22.8  | 15.7  | 10.7  | 3.1  | (27) |
| (28) |                                                                   | MCDWB       | 9.0   | 14.0  | 20.2  | 25.5  | 25.9  | 28.2  | 30.2  | 29.4  | 26.6  | 20.0  | 14.4  | 7.7   | 7.7  | (28) |
| (29) |                                                                   | 2%          | WB    | 0.5   | 4.3   | 9.3   | 14.4  | 17.0  | 20.3  | 23.2  | 23.0  | 20.2  | 14.3  | 9.0   | 1.7  | (29) |
| (30) |                                                                   | MCDWB       | 6.1   | 10.4  | 18.6  | 23.0  | 25.2  | 27.1  | 28.8  | 27.8  | 25.0  | 18.9  | 14.6  | 6.7   | 6.7  | (30) |
| (31) |                                                                   | 5%          | WB    | -0.8  | 3.2   | 7.9   | 13.0  | 16.1  | 19.4  | 22.5  | 22.2  | 18.6  | 13.2  | 7.5   | 0.5  | (31) |
| (32) |                                                                   | MCDWB       | 4.0   | 8.2   | 15.9  | 21.6  | 23.8  | 26.0  | 27.7  | 26.9  | 23.6  | 17.5  | 12.3  | 4.8   | 4.8  | (32) |
| (33) |                                                                   | 10%         | WB    | -2.0  | 2.0   | 6.6   | 11.7  | 15.2  | 18.6  | 21.7  | 21.4  | 17.4  | 12.2  | 6.1   | -0.6 | (33) |
| (34) |                                                                   | MCDWB       | 1.9   | 7.0   | 14.1  | 19.3  | 22.8  | 25.4  | 26.7  | 26.0  | 21.9  | 16.6  | 10.7  | 3.2   | 3.2  | (34) |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 11.7  | 11.8  | 12.4  | 13.0  | 13.0  | 12.4  | 10.3  | 9.8   | 10.0  | 11.4  | 11.2  | 11.0  | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 15.2  | 15.6  | 16.7  | 17.2  | 17.0  | 16.0  | 13.9  | 12.8  | 13.9  | 15.3  | 15.6  | 14.1 | (36) |
| (37) |                                                                   | MCWBR       | 10.0  | 9.0   | 8.0   | 7.0   | 6.4   | 4.9   | 3.9   | 4.1   | 5.1   | 7.0   | 8.4   | 8.9   | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 14.3  | 13.4  | 15.4  | 13.9  | 13.1  | 11.7  | 9.9   | 9.7   | 10.4  | 11.6  | 13.1  | 12.8 | (38) |
| (39) |                                                                   | MCWBR       | 9.6   | 7.9   | 7.8   | 6.3   | 5.5   | 4.4   | 3.6   | 3.8   | 4.5   | 5.8   | 7.6   | 8.4   | 8.4  | (39) |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.386 | 0.466 | 0.496 | 0.521 | 0.526 | 0.572 | 0.568 | 0.591 | 0.536 | 0.494 | 0.409 | 0.362 | (40) |      |
| (41) |                                                                   | taud        | 2.004 | 1.829 | 1.809 | 1.794 | 1.851 | 1.799 | 1.876 | 1.799 | 1.874 | 1.889 | 2.011 | 2.092 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 763   | 738   | 763   | 771   | 775   | 738   | 737   | 707   | 720   | 703   | 731   | 765   | (42) |      |
| (43) |                                                                   | Edh at Noon | 132   | 179   | 200   | 214   | 205   | 216   | 199   | 210   | 185   | 166   | 130   | 113   | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 2.80  | 3.46  | 4.68  | 5.49  | 6.00  | 5.92  | 5.43  | 4.86  | 3.92  | 3.47  | 2.93  | 2.57  | (44) |      |
| (45) |                                                                   | RadStd      | 0.27  | 0.39  | 0.40  | 0.44  | 0.41  | 0.38  | 0.52  | 0.40  | 0.38  | 0.48  | 0.40  | 0.25  | (45) |      |

## Historical Trends

|      |                        | DBAvg |     | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |
|------|------------------------|-------|-----|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|
|      |                        |       |     | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |
|      |                        | (d)   | (e) | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)     | (m)     |         |      |
| (46) | Station Only           | -0.36 | N/A | N/A     | N/A    | -0.48   | N/A   | N/A   | N/A         | N/A     | -130    | -88     | (46) |
| (47) | Regional (4 neighbors) | N/A   | N/A | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | -60     | (47) |

Nomenclature: See separate page